

Optimization in Public Transport

– Exercise Sheet 07

due Monday, June 1 at 10:00

Exercise 1 – Greedy heuristics for (LP1) (4 points)

Generic Greedy Heuristic for Line Planning

Input: PTN, line pool $\mathcal{L}_0$, lower edge frequencies $f_e^{\min} \in \mathbb{N}_0$ for all $e \in E$

Output: A feasible line concept $(\mathcal{L}, f)$ if one exists.

Step 1: $\mathcal{L} := \emptyset$, $f_l := 0$ for all $l \in \mathcal{L}_0$.

Step 2: If $f_e^{\min} = 0$ for all $e \in E$, **stop**.

Output: Feasible line concept $(\mathcal{L}, f)$.

Step 3: Choose a line $l \in \mathcal{L}'$ such that $g(l)$ is minimal.

If no such line exists, **stop**.

Output: No feasible line concept found.

Step 4: Let $\mathcal{L} := \mathcal{L} \cup \{l\}$, increase frequency of line l by $f(l)$, i.e., $f_l := f_l + f(l)$.

For all $e \in l$, update $f_e^{\min} := \max\{f_e^{\min} - f(l), 0\}$.

Go to Step 2.

Consider the generic greedy heuristic for line planning given above.

- (a) Show that the following two heuristics for (LP1) (also described in the lecture notes) find a feasible solution if one exists:

In *Heuristic 7*, we get

Step 3: $g(l) := \frac{\text{cost}_l}{\text{Length}_l}$, with $\mathcal{L}' := \mathcal{L}_0 \setminus \mathcal{L}$

Step 4: $f(l) := \max_{e \in l} f_e^{\min}$.

In *Heuristic 8*, we get

Step 3: $g(l) := \frac{\text{cost}_l}{|\{e' \in l : f_{e'}^{\min} > 0\}|}$ with $\mathcal{L}' := \{l \in \mathcal{L}_0 : e'' \in l\}$ for an edge $e'' \in E$ such that $f_{e''}^{\min}$ is maximal.

Step 4: $f(l) := \min_{e \in l : f_e^{\min} > 0} f_e^{\min}$.

- (b) Define your own greedy heuristic by specifying Step 3 and Step 4 of the generic heuristic. Make sure and prove that your heuristic also finds a feasible solution if one exists.

Exercise 2 – Special cases of (LP2) (4 points)

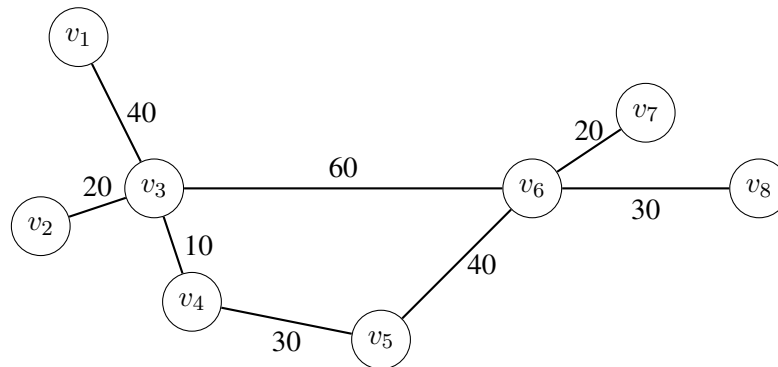
- (a) Show that (LP1) is a special case of (LP2).
- (b) Simplify (LP2) for planning bus lines. (Note that $c_t^{\max} = 1$ for buses.)

Exercise 3 – Programming: greedy heuristics for (LP1) (4+4+2+2 points)

You have time until 08.06. to submit your solutions for this task. You can use any programming language you like (Python is preferred), but pay attention to readability – this includes a reasonable number of comments.

Before we can use our models, we usually need to convert the data into a suitable format. In Section 1.2, we introduced traffic loads. Now, we want to determine these traffic loads for an instance of the PTN "toy". Note, that the graph in task (a) corresponds to the data file "toy", but has different traveling times! The solution of part (a) is not necessary to solve the subsequent parts of this exercise.

- (a) Given the following public transport network with traveling times in minutes for each edge



and the following OD-matrix

$$A = \begin{pmatrix} & v_1 & v_2 & v_3 & v_4 & v_5 & v_6 & v_7 & v_8 \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \\ v_7 \\ v_8 \end{matrix} & \begin{matrix} 0 & 5 & 15 & 10 & 60 & 15 & 4 & 3 \\ - & 0 & 15 & 10 & 60 & 15 & 6 & 1 \\ - & - & 0 & 5 & 80 & 10 & 10 & 10 \\ - & - & - & 0 & 50 & 15 & 15 & 5 \\ - & - & - & - & 0 & 60 & 20 & 25 \\ - & - & - & - & - & 0 & 30 & 20 \\ - & - & - & - & - & - & 0 & 10 \\ - & - & - & - & - & - & - & 0 \end{matrix} \end{pmatrix}.$$

Determine the traffic loads c_e for each edge under the assumption that all passengers travel along shortest paths. Furthermore, choose frequencies $f_e^{\min} = \lceil \frac{c_e}{\text{capacity}} \rceil$ where capacity = 50 is the capacity of each vehicle. Set $f_e^{\max} = f_e^{\min} + 4$. You may solve this exercise by implementing it. Write your solution in a datafile that uses the same format as in "Loads.giv".

For the following tasks, you can use the data files "toy", "grid", "ring", and "bahn-01". For "toy", you may use the constructed datafile from part (a).

- Implement the Heuristics 7 and 8 given in Exercise 1. Test the algorithm on the data provided in OLAT. Which algorithm performs better for which instance?
- Implement your own heuristic from Exercise 1. Test your heuristic on the data provided in OLAT and compare your results to the ones from part (b).
- Write a program to solve (LP1) optimally. Test it on the data provided in OLAT. Did any of the heuristics find an optimal solution?

Please upload a single PDF file per group (with the names on it) in the OLAT course until the due date.